

NICK MARKS

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EXPERIENCE

SWE - AWS Networking

DC Lifecycle and Tooling

📅 09/2025 - Present

📍 Sydney, AUS

- Technical owner of a new automated intake system responding to fiber optic planning signals. Exposed existing failure mechanisms between tools, wrote design docs and developed new workflows to significantly reduce human touchpoints. This system saved 2+ months of engineering time per year and increased speed average intake creation speed by 50x.
- Influenced an initiative across multiple orgs to properly support a change management system for in-flight span planning in DC network planning. Drove consensus amongst different teams to address heavily manual processes, simplify error-prone interactions and clean up poor data quality. Reduced the number of rejected intakes by 40% or ~\$25k per project
- Identified gaps in reuse of existing fiber network and produced a pipeline for aggregating data from multiple sources to surface such spare fiber. Subsequent allocation of spare fiber increased by 15x, reducing fiber deployment times from months to minutes in those cases.

SWE - AWS Relational Database Services

RDS Control Plane Tooling

📅 01/2025 - 09/2025

📍 Sydney, AUS

- Led development of a control plane alarming service with suppression of alarms during high fault-volume instances. This saved RDS/Aurora teams 20+ hours of operational work upon release of fault injection service features to external customers.
- Enhanced a deadline monitoring service receiving 1+ million workflows per hour with "smart" deadlines. This cut or extended workflow timeouts based on historical data to reduce operational load for teams by 32% each week.
- Recognised security vulnerabilities in our org-wide ticket analytics service that compromised access to the service's underlying databases. Proposed and implemented identity-based access methods and rolled out across multiple teams with no loss of access.

EDUCATION

BSc Computer Science and Pure Mathematics

BE Electrical and Computer Systems Engineering

Monash University (WAM: 87.8%, GPA: 3.88/4.00)

📅 03/2020 - 06/2024

📍 Melbourne, VIC

Semester Abroad

Purdue University (WAM: 92.1%, GPA: 4.00/4.00)

📅 01/2023 - 05/2023

📍 West Lafayette, IN

TECHNICAL SKILLS

- **Languages:** Java, C/C++, Go, TypeScript, Python, SQL (Postgres), C#
- **Frameworks:** Node.js, Django, Astro, Vue, NextJS, JUnit, AWS CDK, React
- **AI and HPC:** Pytorch, TensorFlow, OpenMP, CUDA, CMake, Docker, Pandas, Numpy, Bash, Linux
- **Amazon Web Services:** EC2, Lambda, S3, RDS/Aurora, DynamoDB, IAM, API Gateway, Redshift, SQS, SES

COMPETITIONS & RESEARCH

Autonomous Navigation through Offline Deep Learning

📅 07/2023 - 06/2024

- Researched applications of offline deep learning techniques to autonomously navigate a Penguin Pi bot around varying complexities of road tracks.
- Crafted pipeline for training and deploying a model trained via Implicit Q-Learning, using tools like regression analysis and grid-search to optimise trained experts.

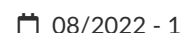
Rocket Guidance Systems at [Purdue Orbital](#)



📅 01/2023 - 05/2023

- Produced control systems on Purdue Orbital's guidance and navigation team for rockets launched by the team in the middle of 2023 as part of Project Icarus.
- Crafted state models and Simulink implementations of rocket attitude control and implemented a real-time operating system to interface between the control system, onboard sensors and diagnostic systems.

Robot-assisted Drone Navigation at [Monash DeepNeuron](#)



📅 08/2022 - 11/2022

- Developed a behavioral cloning-based approach to train a AI-assisted drone navigation system capable of flying point-to-point within simulation.
- Evaluated the model performance in comparison to a reinforcement-learning approach on a dataset generated by human participants in a small-scale trial (~30 participants).

CO2 Conversion Project Lead

[Monash Carbon Capture and Conversion](#)



📅 08/2021 - 12/2023

- Founded a large-scale research project on the electrochemical reduction of CO2 into fuels in a novel, catalysed flow cell environment.
- Our team received US\$250,000 as a top 23 university finalist worldwide in the [XPRIZE Carbon Removal Competition](#).